

RESEARCH INTERESTS

Numerical analysis; approximation theory and applied functional analysis; randomised and numerical linear algebra

EDUCATION

- **Imperial College London** Oct 2026-Jun 2030
Incoming PhD in Mathematics
◦ Supervised by Nicolas Boullé, Greg Pavliotis and Matthew Colbrook (University of Cambridge). London, UK
- **University of Oxford** Oct 2022-Jun 2026
Master of Mathematics
◦ First Class in all three years, final rank 24/126.
◦ Distinction in Masters, supervised by Yuji Nakatsukasa. Dissertation title: *Fast Approximation of Gram Matrix via CUR*. Oxford, UK
- **Keystone Academy** Graduated Jun 2022
International Baccalaureate Diploma
◦ Final Grade: 43/45 Beijing, China

PUBLICATIONS

C=CONFERENCE, J=JOURNAL S=IN SUBMISSION, T=THESIS

- [S.1] Nakatsukasa, Yuji; Zhang, Yifu. 2025. "Efficient Function Approximation Under Heteroskedastic Noise." arXiv:2508.08683 [math.NA].

RESEARCH EXPERIENCE

- **Numerical Linear Algebra** Oct 2025-Jun 2026
Master's Dissertation supervised by Prof. Yuji Nakatsukasa, University of Oxford 
◦ Developed the $CUR(CUR)^T$ approximation for Gram matrix, provided theoretical error bounds using two-sided sketch-and-solve.
◦ Conducted numerical experiments, demonstrating substantial computational speedups over Nyström approximation.
- **Efficient Function Approximation Under Heteroskedastic Noise** Oct 2024-Jul 2025
Supervised by Prof. Yuji Nakatsukasa, University of Oxford 
◦ Developed HeteroChebtrunc, adapting Chebyshev-based approximation to non-uniform noise variance.
◦ Proved high-probability sup-norm bounds under heteroskedastic sampling (for subgaussian noise).
◦ Consistently lower sup-norm error across noise profiles; $O(N)$ runtime compared with its predecessor's $O(N \log N)$.
◦ Open-sourced MATLAB code.
- **Project in Analytic Number Theory** Jun 2024-Sept 2024
Supervised by Dr. Ofir Gorodetsky
◦ Studied structural links between permutation groups and integers.
◦ Generalised an asymptotic bound of Eberhard–Green–Ford (2016) under Ewens θ measure with $0 < \theta \leq 1$.
◦ Extended key parts of the Ford (2016) framework from the uniform to Ewens θ measure.

HONORS AND AWARDS

- **Scholar (2025, 2024) and Exhibitioner (2023)**
University College, University of Oxford
◦ For outstanding performance in examinations.
- **Shing-Tung Yau High School Mathematics Award (2021)**
Tsinghua University
◦ Global top 10 among 200+ teams and individual entrants for original research paper in coding theory.
- **Keystone Scholarship (2021, 2022)**
Keystone Academy
◦ Full tuition coverage for final HS year and first undergraduate year.

SKILLS

Languages: English (Fluent), French (B1), Mandarin (Native).

Coding: Python (numpy, scipy), MATLAB (Chebfun)

Teaching: Tutoring for mock interviews; maths classes for summer school students (70+ hrs); creation of problem sets and exam papers.

REFERENCES

Available upon request.